Motion through space is a concept we learn about early in life. Although we often don't think of things in such terms, we also learn that we, our friends, our belongings, and so forth all move through time, as well. When we look at a clock or a wristwatch, even while we idly sit and watch TV, the reading on the watch is constantly changing, constantly "moving forward in time." We and everything around us are aging, inevitably passing from one moment in time to the next. In fact, the mathematician Hermann Minkowski, and ultimately Einstein as well, advocated thinking about time as another dimension of the universe—the fourth dimension—in some ways quite similar to the three spatial dimensions in which we find ourselves immersed. Although it sounds abstract, the notion of time as a dimension is actually concrete. When we want to meet someone, we tell them where "in space" we will expect to see them—for instance, the 9th floor of the building on the corner of 53rd Street and 7th Avenue. There are three pieces of information here (9th floor, 53rd Street, 7th Avenue) reflecting a particular location in the three spatial dimensions of the universe. Equally important, however, is our specification of when we expect to meet them—for instance, at 3 P.M. This piece of information tells us where "in time" our meeting will take place. Events are therefore specified by four pieces of information: three in space and one in time. Such data, it is said, specifies the location of the event in space and in time, or in spacetime, for short. In this sense, time is another dimension.

Since this view proclaims that space and time are simply different examples of dimensions, can we speak of an object's speed through time in a manner resembling the concept of its speed through space? We can.

A big clue for how to do this comes from a central piece of information we have already encountered. When an object moves through space relative to us, its clock runs slow compared to ours. That is, the speed of its motion through time slows down. Here's the leap: Einstein proclaimed that all objects in the universe are always traveling through spacetime at one fixed speed—that of light. This is a strange idea; we are used to the notion that objects travel at speeds considerably less than that of light. We have repeatedly emphasized this as the reason relativistic effects are so unfamiliar in the everyday